

CERTIFICATE OF PARTICIPATION



June 22 - July 4th, 2026

THIS CERTIFIES THAT

Thankaswamy Shajin

HAS SUCCESSFULLY PARTICIPATED IN THE
GRAY SCOTT SCHOOL 2026

FOCUSING ON PROGRAMMING AND OPTIMIZATION ON
HETEROGENEOUS ARCHITECTURES,

FOR A TOTAL AMOUNT OF HOURS OF: **51,5 HOURS**

Organized in Annecy and broadcsated online and in 19 satellite sites, by:



Confirmation of Participation

Shajin Thankaswamy

took part in the course

The Gray Scott HPC Summer School – Stuttgart Satellite Site

held at the High-Performance Computing Center Stuttgart (HLRS) from June 22 – July 2, 2026
and attended 51 hours 30 min of lectures and exercises on the following topics:

Day 1

- CPU Architecture
- Vectorization VS Parallelization?
- What is compilation?
- Did you say Unit Tests?

Day 2

- C++ 17/20/23 on CPU
- C++ Advanced Optimization (Pyramid)
- Theory and profiling tools

Day 3

- Fortran 2018 on CPU
- Memory Profiling Tools

Day 4

- Kokkos on CPU
- Computation and digital precision

Day 5

- Python computing on CPU
- Impact of the system on applications performances

Day 6

- C++ 20 Computing with Eve + Kiwaku
- GPU Architecture massively parallel computing
- C++17/20/23 on GPU with NVC++

Day 7

- Python GPU computing
- Python GPU computing: CuPy
- Python GPU computing: CuNumeric
- Python GPU computing: JAX

Day 8

- Fortran 2018 on GPU
- Julia ... in Rust ... and C++ ... with pixi?

Day 9

- Kokkos on GPU
- A story about cubic root optimisation in C++ and Fortran

The lecturers and trainers were Pierre Aubert, Hadrien Grasland, Vincent Lafage, Sébastien Valat, David Chamont, Alice Faure, Jean-Marc Colley, Nabil Garroum, Joël Falcou, Thomas Padioleau, Paul Zehner and Juan-José Silva Cuevas.

Stuttgart, July 2, 2026



Tobias Haas
Course Organizer

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